

On the role of coherent structures in turbulent wake evolution

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Coherent structures (CS) have been found to coexist with fine-scale turbulence in a wide range of flows. However, the role of CS in the evolution of the mean flow and their interactions with broadband turbulence have not been clear. In this work, numerical experiments are designed to isolate the effects, or lack thereof, of the CS in an axisymmetric, high-Reynolds-number wake past a circular disk. The disk wake features strong vortex-shedding (VS) structures.

Planar data are taken at a near-wake streamwise location from an unforced, baseline simulation. After manipulation of the flow, the data are fed as inflow conditions into the second-stage simulations that proceed into the far wake. Manipulations corresponding to altering the spectral content of the wake in different ways are performed.

The evolution of the mean and the turbulent statistics, as well as the inter-scale energy transfer, are compared among the modified wakes and the unforced wake. The streamwise decay or regrowth of the CS under different inflow conditions is also tracked. It is found that the CS qualitatively change the downstream evolution of the mean flow, whereas their effects on the spectral energy distribution of the TKE are not as strong, once the small scales have adjusted to the large scales.

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